

OTCSafe

Agentic AI vs. Direct LLM Comparisons

Four identical OTC safety queries were submitted to OTCSafe (agentic pipeline: RxNorm + OpenFDA + Claude), Gemini, and ChatGPT. Each response is evaluated on the same dimensions. The goal is to show how the agentic approach produces meaningfully different output than direct LLM responses, and where those differences lie.

The agent's core claim is that there is a meaningful difference between an LLM's best effort to answer a question based on its training data alone, versus an agentic architecture that grounds the LLM's reasoning in live retrieval from pharmacological databases, calculating real dose figures, and citing verifiable sources produces meaningfully different output from LLM memory alone — particularly for safety-critical decisions where accuracy matters and for keeping humans in the loop with transparent sourcing.

Reading the comparison tables

✓ Agent / LLM did this

Correctly addressed this dimension

~ Partial

Addressed but incomplete or hedged

✗ Missed

Not addressed or incorrect

US medications only · Research prototype · Not medical advice

Example 01

Query

I took 2 DayQuil gel caps at 9am, then 2 Tylenol Extra Strength at 11am, and just drank a Theraflu tea. Is this too much?

Moderate Risk

Summary of agent's response: Flagged acetaminophen overlap with dose calculated from live OpenFDA labels (1,500 mg). Also caught chlorpheniramine + dextromethorphan interaction within DayQuil — not mentioned by either LLM.

Dimension	OTCSafe (Agent)	Gemini	ChatGPT
Acetaminophen overlap caught?	✓ Yes	✓ Yes	✓ Yes
Dose figure (mg)	1,500 mg (live label data)	2,150–2,650 mg (training estimate)	~2,300 mg (training estimate)
Source cited?	✓ OpenFDA URL per product	✗ None	✗ None
Extra interaction found?	✓ Chlorpheniramine + DXM (DayQuil)	✗ Missed	✗ Missed

Agent stronger at	Dose precision from live data. Extra interaction found.
LLMs stronger at	Stronger escalation. Poison control number provided.

Verdict	The agent does better on factual accuracy — its 1,500 mg figure comes from live label data, not memory, and more importantly: it caught an intra-product interaction the LLMs missed. Both LLMs win on escalation: they told the user to call poison control, which is more appropriate than 'see a pharmacist'. This can be added/changed in the agent's synthesis prompt without changing the overall architecture, which is a plus for the agentic approach.
----------------	---

Example 02

Query	<i>I'm on an SSRI and I've been sipping Robitussin all day for a cough. My heart feels weird. Did I mess up?</i>
--------------	--

Moderate Risk	Summary of agent's response: Correctly identified serotonin syndrome risk (SSRI + dextromethorphan). Named safe alternatives (plain Mucinex, benzonatate). Cited OpenFDA label for Robitussin.
----------------------	--

Dimension	OTCSafe (Agent)	Gemini	ChatGPT
Serotonin syndrome identified?	✓ Yes, with mechanism	✓ Yes — strongest explanation	✓ Yes
CYP450 enzyme point mentioned?	✗ Not mentioned	✓ Yes — SSRI blocks enzyme for DXM	✗ Not mentioned
Safe alternative named?	✓ Plain Mucinex, benzonatate	✓ Honey, menthol cough drops	✗ Not specific
Escalation urgency	~ See pharmacist	✓ Call poison control, check temp, ER if fever	✓ Poison control number given

Dimension	OTCSafe (Agent)	Gemini	ChatGPT
Source cited?	✓ OpenFDA label for Robitussin	✗ None	✗ None

Agent stronger at	Best safe alternatives named. Source cited.
LLMs stronger at	Stronger escalation.

Verdict	Gemini connected the user's 'heart feels weird' symptom directly to serotonin toxicity. The agent correctly identified the risk and gave the best safe alternatives, but its escalation was soft for someone reporting active symptoms. All three caught the core risk. The right sources for informed decision making were only cited by the agent, which is a major advantage for transparency and trust especially when it comes to health decisions. The LLMs' lack of sourcing is a known issue that can lead to misinformation or keep humans out of the loop for important health information.
----------------	---

Example 03

Query	<i>I took a Benadryl this morning for allergies, using Dramamine for a boat trip later, and plan to take ZzzQuil tonight. Is that three different medications?</i>
--------------	--

Moderate Risk	Summary of agent's response: Correctly identified all three as diphenhydramine equivalents (including Dramamine's dimenhydrinate → diphenhydramine conversion). Flagged duplicate dosing. Cited RxNorm + OpenFDA for each product.
----------------------	--

Dimension	OTCSafe (Agent)	Gemini	ChatGPT
Duplicate ingredient identified?	✓ All three = diphenhydramine	✓ Yes — breakdown table included	✓ Yes — explicitly stated
Dramamine mechanism explained?	✓ Dimenhydrinate converts to DPH	✓ Called it '~50% diphenhydramine'	✓ Noted it's 'closely related'
Anticholinergic risk named?	~ Named drowsiness, falls	✓ Named anticholinergic toxicity explicitly	✓ Named anticholinergic effects
Older adult risk flagged?	✓ Yes — falls, memory	✓ Yes — confusion, hallucinations	✗ Not flagged
Boat trip specific advice?	✗ Not addressed	✓ 'Stick with Dramamine for boat'	~ Brief mention only
Source cited?	✓ RxNorm + OpenFDA URL per product	✗ None	✗ None

Agent stronger at	Full source citations. Older adult risk flagged.
LLMs stronger at	Gemini: scenario-specific advice given.

Verdict	All three performed well on the core question — correctly identifying that these are the same drug. The agent's main advantage was structured sourcing; its main gap was not addressing the boat trip context.
----------------	--

Example 04

Query	<i>I take fish oil, St. John's Wort, melatonin, and I'm on birth control. Should I be worried?</i>
--------------	--

High Risk	Summary of agent's response: Correctly flagged HIGH risk. Identified three distinct interactions: St. John's Wort → birth control failure (CYP3A4), St. John's Wort + fish oil → bleeding risk, St. John's Wort → reduced melatonin effectiveness. Advised backup contraception immediately.
------------------	--

Dimension	OTCSafe (Agent)	Gemini	ChatGPT
Birth control interaction caught?	✓ Yes — CYP3A4 mechanism	✓ Yes — strongest explanation	✓ Yes
Risk level assigned correctly?	✓ HIGH — correct for pregnancy risk	✓ Labeled high for this interaction	~ Not explicitly rated
Fish oil interaction identified?	✓ Yes — bleeding risk with SJW	~ Noted blood thinning only	~ Brief mention only
Melatonin + SJW noted?	✓ Yes — SJW reduces melatonin effect	✓ Yes — sedation and serotonin risk	~ Drowsiness only noted
Backup contraception advised?	✓ Yes — stop SJW, use condoms, 1–2 weeks	✓ Yes — 4 weeks after stopping	✓ Yes — 2 weeks after stopping
Source cited?	✓ RxNorm + LLM reasoning cited	✗ None	✗ None

Agent stronger at	Three distinct interactions found. HIGH risk correctly assigned. All sources cited.
LLMs stronger at	Gemini: generated summary table.

Verdict The agent is the only system that flagged all three distinct interactions (birth control failure, bleeding risk, reduced melatonin efficacy), assigned the correct HIGH risk level, and cited sources. Gemini gave a useful summary table. This is one of the clearest example of where multi-interaction detection via the LLM reasoning step outperforms direct LLM responses.

Overall Summary

	OTCSafe	Gemini	ChatGPT
Dose figures from live data	✓	✗	✗
All interactions cited to source	✓	✗	✗
Multi-interaction detection	✓ (3–4 per query)	✓ (1–2)	✓ (1–2)
Escalation urgency	~ (too soft)	✓	✓
Poison control number given	✗	✓	✓
Older adult risk flagged	✓	✓	✗
Safe alternative drugs named	✓	✓	~ (partial)

The agent's consistent advantages are factual grounding (live data, cited sources) and multi-interaction detection. Its consistent gap is escalation tone — 'speak with a pharmacist' is often too soft. This is addressable in the synthesis prompt without changing the architecture.